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Assignment Module 7

<https://github.com/alemartuw/DBFoundationsAMartin/tree/DBFoundations-Module7>

Assignment 7

1. Explain when you would use a SQL UDF.
2. Explain the differences between Scalar, Inline, and Multi-Statement Functions.

1. Introduction

In this module, we will learn how to use a function, the types of functions and when to use each one of them.

2. Using a SQL UDF

A user Defined function are custom functions created by the user to be reused. It can be used to calculate values, perform transformations, or apply complex logics.

3. Differences between Scalar, Inline, and Multi-Statement Functions

A **scalar function** returns a single value. It is used when you need to perform an operation that computes one value based on the input parameters. Example:

```
CREATE FUNCTION dbo.ftable (@value1 FLOAT, @value2 FLOAT)
RETURNS FLOAT AS
BEGIN
    RETURN @value1 * @value2
END
```

An **inline or table valued function (TVF)** returns a table as a result. The query is written in the RETURN statement. Only one SELECT is used in the function. Example:

```
CREATE FUNCTION dbo.ftable1 (@value1 INT)
RETURNS TABLE
AS
RETURN
(
    SELECT column1, column2
    FROM table
    WHERE column1 = @value1
);
```

A **multi-statement function or Multi-statement table-valued function (TVF)** also returns a table, but it allows you to include multiple SQL statements and procedural logic. It allows for loops and conditional statements. Example:

```
CREATE FUNCTION dbo. Ftable1 (@value1 INT)
RETURNS @tableVar TABLE (column1 INT, column2 VARCHAR(100))
AS
BEGIN
```

```
INSERT INTO @tableVar (column1, column2)
SELECT column1, column2, FROM table2 WHERE value1 = @value1 ; -- Additional logic can be added here,
such as filtering, sorting, etc.
RETURN;
END
```

Summary

I learned how use Scalar, Inline, and Multi-Statement Functions. I also learned that when using inline table function you don't use BEGIN and END.

References:

1. Class Notes
2. ChatGPT
3. Quora